

Verhelende uitwerking van §3.8 2c

HTJ

$$y(t) = \mathcal{L}^{-1}\left(\frac{2s+1}{4s^2+16s+32}\right) = \frac{2}{4} \mathcal{L}^{-1}\left(\frac{s+1/2}{s^2+4s+8}\right)$$

Let op: discriminant is negatief, dus kwadraat-afspitten.

$$= \frac{1}{2} \mathcal{L}^{-1}\left(\frac{s+1/2}{(s+2)^2+4}\right) = \frac{1}{2} \mathcal{L}^{-1}\left(\frac{s+2}{(s+2)^2+4} - \frac{3/2}{(s+2)^2+4}\right)$$

$$= \frac{1}{2} \left[\mathcal{L}^{-1}\left(\frac{s+2}{(s+2)^2+4}\right) - \mathcal{L}^{-1}\left(\frac{3/2}{(s+2)^2+4}\right) \right]$$

$$= \frac{1}{2} \cdot e^{-2t} \cdot \left(\cos(2t) - \frac{3}{4} \sin(2t) \right)$$